MOODY'S

Understanding GCorr 2023 Corporate

Abstract

The Moody's Global Correlation Model (GCorr®) is a multi-factor model that assesses asset correlations. It is used within Moody's credit portfolio products. Simulation engines in these products use GCorr to randomly draw correlated asset returns and subsequently calculate the corresponding values of financial instruments at a future time horizon.

GCorr Corporate is a forward-looking, multi-factor model that assesses asset correlations of publicly-traded firms. The model is re-estimated each year, incorporating the most recent dynamics of firms' financial situations, as well as general economic conditions. This paper summarizes and analyzes the most recent parameter estimates (GCorr 2023 Corporate) and discusses the impact of these estimates on portfolio risk measures.

The dataset used for estimation of the 2023 model contains additional 10 months of firms' asset returns and covers 53,436 firms globally.

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1. Introduction

A credit portfolio model is an essential input of credit risk measurement and management for most financial institutions. Measuring the riskiness of single instruments does not suffice — a portfolio of instruments with low stand-alone default probabilities but high correlations may still pose a substantial risk of large losses. Therefore, it is important to understand how correlations among various instruments affect portfolio credit risk. The Moody's portfolio products employ GCorr® to determine asset correlations among borrowers. The model is used by simulation engines in these products to randomly draw correlated asset returns and subsequently calculate the corresponding values of financial instruments at a future time horizon.

Developed by Moody's, GCorr Corporate is a forward-looking, multi-factor model for asset correlations of publicly-traded firms. The model is re-estimated each year, incorporating the most recent dynamics of firms' financial situations as well as general economic conditions. This paper summarizes and analyzes the most recent estimates of the GCorr Corporate model: GCorr 2023 Corporate. In addition, the paper discusses the impact of using these GCorr estimates on portfolio risk measures. We assume that readers are familiar with the GCorr model. For a detailed description of the model and the estimation process, see Huang et al. (2015).

There are many drivers of change in observed correlations over time. Globalization and structural economic changes, such as the introduction of the Euro in 1999, affect long-term trends and levels of observed correlations. As data shows, correlations are also subject to shorter-term fluctuations, which tend to be mean-reverting. In some cases, the increase in observed correlations above a long-term trend coincides with an economic downturn. The 2008-2009 financial crisis and the COVID-19 pandemic in 2020 are examples of this pattern. The GCorr Corporate model of asset correlations incorporates all these features of observed correlation dynamics. The Moody's paper "Validation of GCorr 2023 Corporate"[1] discusses correlation behavior over time in more detail, assesses performance of the forward-looking feature of GCorr Corporate, and examines the differences between the forward-looking GCorr Corporate correlations and empirical correlations observed over a recent period.

In this study, we discuss the updates to the 2023 model, which include extra 10 months of asset return data. We also analyze correlation patterns in the four most recent correlation estimates — GCorr 2023 Corporate, GCorr 2022 Corporate, GCorr 2021 Corporate, and GCorr 2020 Corporate. For information about earlier models, see Xiao and Lanfranconi (2023) and the references contained therein.

In addition to GCorr Corporate, the GCorr framework also includes GCorr CRE (a model for asset correlations of U.S., Canadian and European commercial real estate borrowers), GCorr Retail (a model for asset correlations of U.S., Canadian and European retail borrowers), and GCorr Macro (a model for correlations between credit factors and macro variables with a variety of applications). GCorr components are linked via a common set of factors. For details regarding GCorr CRE, GCorr Retail, and GCorr Macro, see Pei and Lanfranconi (2024), Pei and Lanfranconi (2022), and Huang et al. (2017). The GCorr framework also includes models that assess asset correlations for sovereign nations and small to medium-sized firms. For information regarding these models, see the Moody's white papers Lkhagvajav et al. (2024) and Pei et al. (2024).

[1] See Lkhagvajav et al. (2024).

2. Methodology

To begin this study, we seek to understand the differences between the GCorr® correlation models by examining the correlation estimates across four correlation models — GCorr 2023 Corporate, GCorr 2022 Corporate, GCorr 2021 Corporate, and GCorr 2020 Corporate. We provide the medians of pairwise correlations for countries and industries. These medians indicate whether a portfolio will undergo significant changes in correlations given its country and industry exposures. New estimates from GCorr 2023 Corporate might have more impact on risk measures of portfolios with such exposures.

Next, we examine R-squared estimates across the four most recent correlation models. R-squared values measure the level of systematic risk for each firm. These values are defined as the square of the correlation between the asset return of a firm and its systematic factor, specified by the country and industry factors to which the firm is exposed. R-squared values represent an important determinant of asset correlations; therefore, examining the changes in R-squared values across countries and industries can provide additional insights.

Following R-squared analysis, we examine systematic factor correlations across the recent models. Systematic factor correlations determine the level of correlation between the different countries and industries and are another important determinant of asset correlations. The asset correlations are driven by the firms' R-squared values as well as their systematic factor correlations.

We also examine the impact of new correlation estimates on 16 portfolios. Although the portfolios are synthetic, they are designed to represent realistic investments. We calculate the unexpected loss (UL) and the economic capital with respect to expected loss (EL) for each of the portfolios using the standard RiskFrontier™ configurations and GCorr 2022 Corporate. Subsequently, we repeat the exercise with GCorr 2023 Corporate, while keeping the rest of the settings and input data constant to isolate the effects of the new correlation estimates. Changes in the portfolio UL reflect the impact of the new correlation model on the variance risk, or the volatility of the portfolio losses. Changes in the economic capital allow us to see how the new correlations affect the portfolio tail risk.

3. Asset Correlation Analysis

To gain more insight into the impact of the new correlation estimates on portfolio risk measures, we compare asset correlations and R-squared values obtained from the correlation models. In this section, we summarize the GCorr® asset correlations across countries and industries, and observe how they have evolved over time. We cover 53,436 publicly-traded firms in our database. To calculate every possible pairwise asset correlation among the firms would be time consuming and computationally expensive as the total number of pairwise correlations is more than 500 million. Considering this, we focus on correlations in random samples; results in this study should be taken as indicative.

We randomly select 5,000 firms from our database and calculate the GCorr asset correlations of all possible combinations (*for example*, more than 12 million correlation values). To gauge the impact of firm size on asset correlations, we construct three groups based on the firms' market value of assets. Specifically, we define firms with market asset values at the end of March 2023 greater than $5 billion as the large-sized group, between $500 million and $5 billion as the medium-sized group, and under $500 million as the small-sized group. We then randomly select 500 firms from each group, and label them as Large500, Medium500, and Small500, respectively. We employ six correlation models (GCorr 2018 — GCorr 2023) to estimate all possible asset correlations among the firms in these samples and report the characteristics of the distributions.

Table 1 summarizes GCorr asset correlations based on the random sample of 5,000 firms.

Table 1 Correlation Summary Statistics: 5,000 Firms

MODEL	MEAN	STD. DEV.	MINIMUM	25TH PCT	MEDIAN	75TH PCT	MAXIMUM
GCorr 2018 Corporate	8.8%	5.8%	0.1%	5.1%	7.3%	10.6%	65.0%
GCorr 2019 Corporate	8.5%	5.6%	0.1%	5.0%	7.1%	10.3%	65.0%
GCorr 2020 Corporate	9.6%	5.8%	0.1%	5.7%	8.1%	11.7%	65.0%
GCorr 2021 Corporate	8.6%	5.3%	0.2%	5.1%	7.2%	10.4%	65.0%
GCorr 2022 Corporate	8.3%	5.0%	0.3%	5.0%	7.0%	10.1%	65.0%
GCorr 2023 Corporate	8.4%	5.0%	0.2%	5.1%	7.1%	10.2%	65.0%

For this random sample of 5,000 firms, the mean of GCorr 2018 Corporate correlations is 8.8%. This mean subsequently changes to 8.5% in GCorr 2019 Corporate, 9.6% in GCorr 2020 Corporate, 8.6% in GCorr 2021, 8.3% in GCorr 2022 Corporate, and 8.4% in GCorr 2023 Corporate. GCorr 2023 Corporate provides slightly higher correlations when compared to GCorr 2022 Corporate for this particular sample.

Table 2, Table 3, and Table 4 present statistics for the estimated asset correlations in the Large500, Medium500, and Small500 samples, respectively. Focusing on the changes from GCorr 2022 Corporate to GCorr 2023 Corporate, the tables show nearly identical mean and median correlations for the Medium500 and the Small500, and a slight increase in the correlations for the Large500 group.

Table 2 Correlation Summary Statistics: Large500 Sample

MODEL	MEAN	STD. DEV.	MINIMUM	25TH PCT	MEDIAN	75TH PCT	MAXIMUM
GCorr 2018 Corporate	16.2%	9.4%	0.9%	9.3%	13.9%	20.8%	65.0%
GCorr 2019 Corporate	15.8%	9.4%	0.8%	9.0%	13.4%	20.1%	65.0%
GCorr 2020 Corporate	17.7%	9.8%	1.3%	10.4%	15.3%	22.6%	65.0%
GCorr 2021 Corporate	16.5%	9.6%	1.1%	9.6%	14.1%	21.1%	65.0%
GCorr 2022 Corporate	15.8%	9.3%	1.0%	9.0%	13.4%	20.3%	65.0%
GCorr 2023 Corporate	16.0%	9.4%	0.9%	9.1%	13.6%	20.7%	65.0%

Table 3 Correlation Summary Statistics: Medium500 Sample

MODEL	MEAN	STD. DEV.	MINIMUM	25TH PCT	MEDIAN	75TH PCT	MAXIMUM
GCorr 2018 Corporate	12.2%	9.0%	0.8%	6.8%	9.8%	14.2%	62.4%
GCorr 2019 Corporate	11.6%	8.2%	0.7%	6.5%	9.4%	13.7%	59.2%
GCorr 2020 Corporate	12.8%	7.8%	1.0%	7.5%	10.7%	15.3%	60.2%
GCorr 2021 Corporate	11.3%	6.9%	0.6%	6.6%	9.5%	13.8%	60.3%
GCorr 2022 Corporate	10.4%	6.2%	0.6%	6.1%	8.8%	12.9%	60.2%
GCorr 2023 Corporate	10.4%	6.1%	0.7%	6.0%	8.8%	13.0%	57.8%

Table 4 Correlation Summary Statistics: Small500 Sample

MODEL	MEAN	STD. DEV.	MINIMUM	25TH PCT	MEDIAN	75TH PCT	MAXIMUM
GCorr 2018 Corporate	6.7%	3.7%	0.8%	4.3%	5.8%	8.0%	56.3%
GCorr 2019 Corporate	6.5%	3.6%	0.7%	4.2%	5.7%	7.7%	57.8%
GCorr 2020 Corporate	7.4%	4.0%	0.8%	4.8%	6.5%	8.9%	58.7%
GCorr 2021 Corporate	6.6%	3.4%	0.6%	4.3%	5.7%	7.8%	56.4%
GCorr 2022 Corporate	6.6%	3.4%	0.6%	4.4%	5.8%	7.9%	59.9%
GCorr 2023 Corporate	6.6%	3.3%	0.8%	4.4%	5.7%	7.9%	60.2%

ASSET CORRELATIONS BY COUNTRY

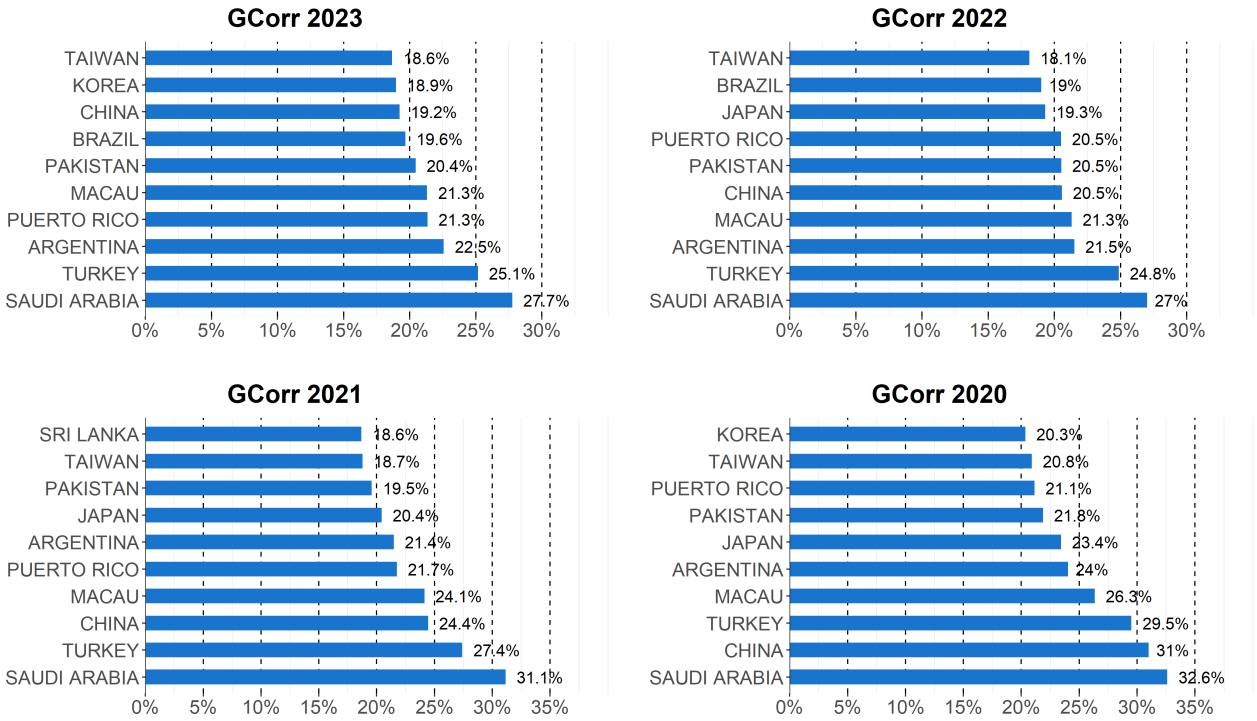
This section analyzes GCorr asset correlations by country. We choose a representative sample of 200 randomly selected firms from each country. When there are fewer than 200 firms in a country, we select all the firms for that country.

Figure 1 and Figure 2 plot the top median within-country correlations and the top differences in the median within-country correlations between two models. We calculate these medians from the pairwise asset correlations of all firms belonging to the representative sample for the country.[2] The countries that appear in the top-ten list are consistent across the models. All of the top-ten countries in GCorr 2022 Corporate except Japan remain among the top ten for GCorr 2023 Corporate as well.

Figure 1 shows the top-ten countries with the highest median within-country correlations.

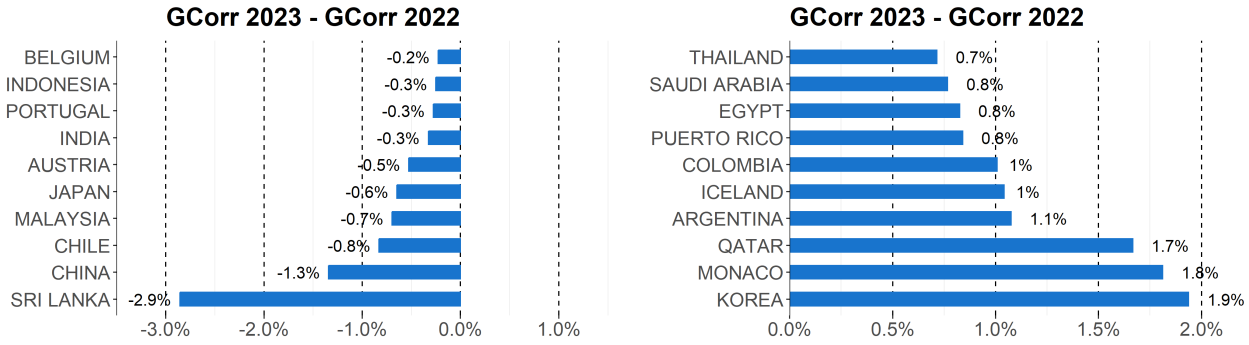
[2]The results reported in this paper regarding GCorr 2022 Corporate, GCorr 2021 Corporate, and GCorr 2020 Corporate may not necessarily be the same as those reported in the papers *Understanding GCorr 2022 Corporate*, *Understanding GCorr 2021 Corporate*, and *Understanding GCorr 2020 Corporate*. For each country, we choose 200 firms randomly to construct a representative sample. Due to the random sampling, median correlation can be slightly different from that reported in the previous papers.

Figure 1 Top 10 median correlations within a country based on GCorr 2023 Corporate, GCorr 2022 Corporate, GCorr 2021 Corporate, and GCorr 2020 Corporate.



The following figure (Figure 2) shows the countries for which the within-country median asset correlation has increased or decreased the most, from GCorr 2022 Corporate to GCorr 2023 Corporate.

Figure 2 Top 10 negative and positive differences in median within-country correlations: GCorr 2023 Corporate — GCorr 2022 Corporate.



ASSET CORRELATIONS BY INDUSTRY

This section examines asset correlations by industry. Similar to country correlations, we work with a representative sample for each industry by randomly selecting 200 firms. If there are fewer than 200 firms in one industry, we select all the firms in the industry.

Figure 3 and Figure 4 present the top 10 industries with the highest median within-industry correlations and the largest changes in the median within-industry correlations. We calculate the median values from the pairwise correlations of all firms belonging to the same representative industry sample. The industries that

appear in the top 10 list are mostly consistent across the models. All of the top 10 industries in GCorr 2022 Corporate remain among the top 10 for GCorr 2023 Corporate as well.

Figure 3 shows the top 10 industries with the highest median within-industry correlations.

Figure 3 Top 10 median correlations within a industry based on GCorr 2023 Corporate, GCorr 2022 Corporate, GCorr 2021 Corporate, and GCorr 2020 Corporate.

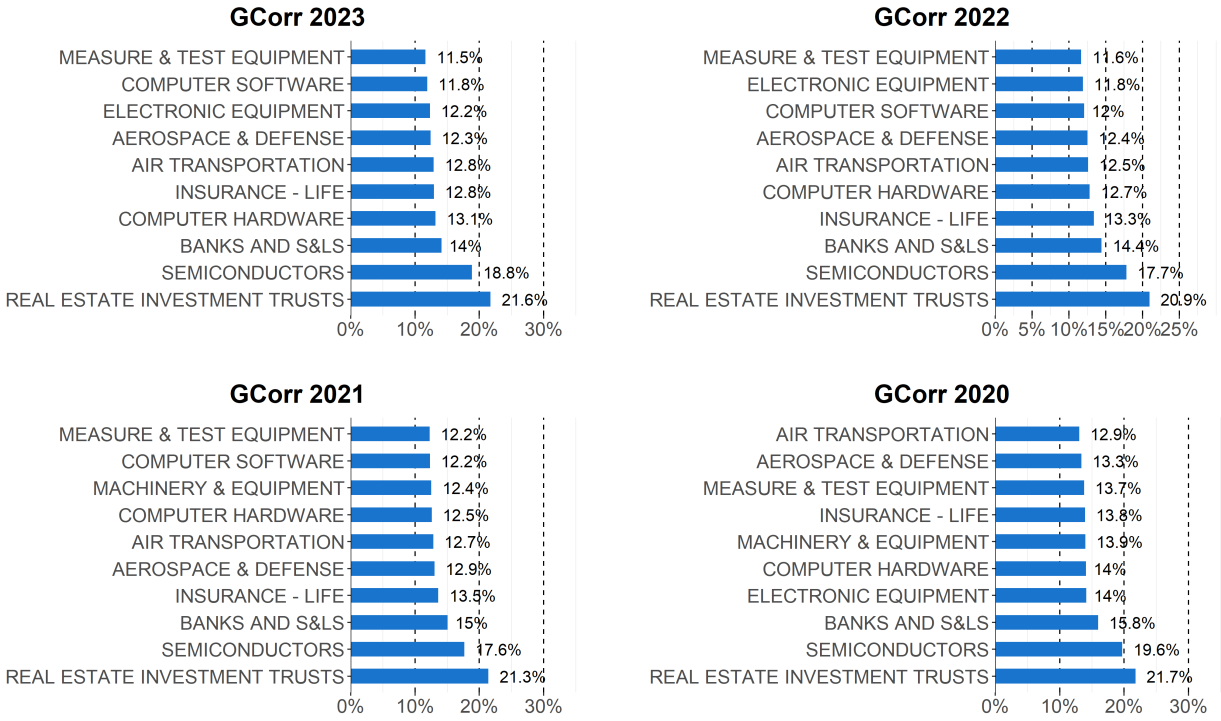
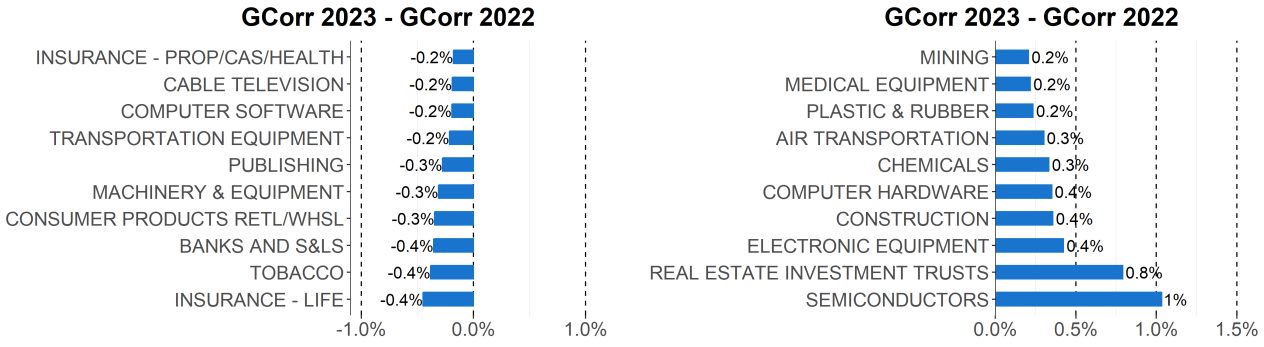


Figure 4 shows the industries for which the within-industry median asset correlation has the most decrease or increase from GCorr 2022 Corporate to GCorr 2023 Corporate. Semiconductors is the industry with the highest increase from GCorr 2022 to GCorr 2023, potentially driven by the steep decline of global semiconductors industry sales in 2023.

Figure 4 Top 10 negative and positive differences in median within-industry correlations: GCorr 2023 Corporate — GCorr 2022 Corporate.



4. R-squared Analysis

In the previous section, we analyzed the differences in the asset correlation estimates produced by the four GCorr® correlation models. Since a firm's R-squared value (the proportion of the firm's risk attributable to the systematic or the economy-wide risk) is one of the drivers of asset correlations, we also examine the distribution of R-squared values in GCorr 2023 Corporate and GCorr 2022 Corporate.

We first provide a brief overview of how R-squared values affect asset correlations. Consider the following equation.

$$r_k = \beta_k \phi_k + \epsilon_k \tag{1}$$

Here, β_k measures the sensitivity of firm k's asset return to its systematic factor ϕ_k, and ϵ_k represents the idiosyncratic portion of the return. The R-squared value measures the proportion of the firm k's risk explained by the systematic factor. This statement can be expressed mathematically as follows:

$$\begin{aligned} R^2 &= \frac{\beta_k^2 \sigma_{\phi_k}^2}{\sigma_k^2} \\ &= \frac{\beta_k^2 \sigma_{\phi_k}^2}{\beta_k^2 \sigma_{\phi_k}^2 + \sigma_{\epsilon_k}^2} \end{aligned} \tag{2}$$

where σ_k^2, $\sigma_{\phi_k}^2$, and $\sigma_{\epsilon_k}^2$ are the unconditional variances of r_k, ϕ_k, and ϵ_k, respectively.

Now, consider the asset returns for firm j and firm k:

$$\begin{aligned} r_j &= \beta_j \phi_j + \epsilon_j \\ r_k &= \beta_k \phi_k + \epsilon_k \end{aligned} \tag{3}$$

The correlation coefficient between r_j and r_k is obtained as follows:

$$\rho_{j,k} = \frac{\beta_j \beta_k cov(\phi_j, \phi_k)}{\sigma_j \sigma_k} \tag{4}$$

Note, the correlation is simply a scaled covariance for which the denominator is the product of the standard deviations of the individual asset returns. Simple algebraic manipulations show that the asset correlation between two firms can be written as a product of the correlation between the systematic factors and the square root of the R-squared values of these firms.

$$\begin{aligned} \rho_{j,k} &= \frac{\beta_j \beta_k \sigma_{\phi_j} \sigma_{\phi_k}}{\sigma_j \sigma_k} \frac{cov(\phi_j, \phi_k)}{\sigma_{\phi_j} \sigma_{\phi_k}} \\ &= \left(\frac{\beta_j \sigma_{\phi_j}}{\sigma_j} \right) \left(\frac{\beta_k \sigma_{\phi_k}}{\sigma_k} \right) \rho_{\phi_j \phi_k} \\ &= \sqrt{R_j^2} \sqrt{R_k^2} \rho_{\phi_j \phi_k} \end{aligned} \tag{5}$$

If two firms belong to the same industry and the same country, their systematic factors will be the same, and their asset correlation will become the geometric average of their R-squared values.

Next, we present the R-squared estimates based on the previous correlation models. First, we determine the sample mean, median, and standard deviation for the sample of all public firms in our database. Table

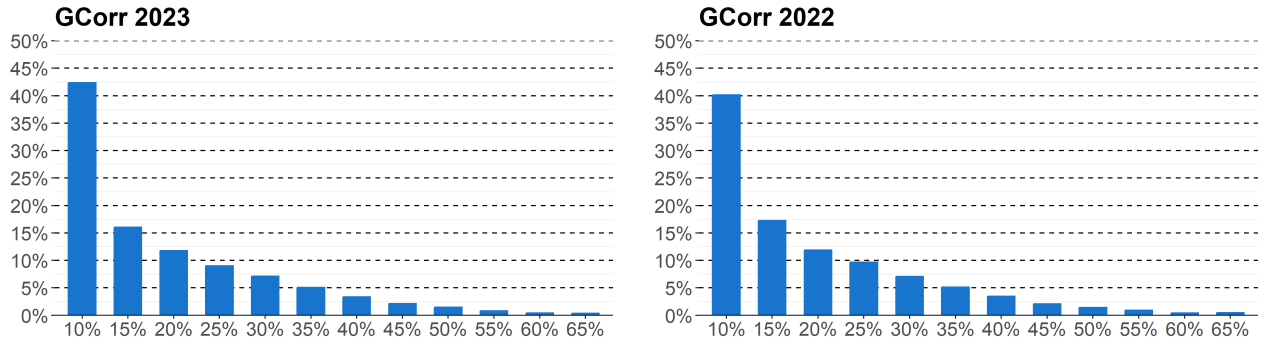
5[3] summarizes some statistics of the R-squared values for GCorr 2023 Corporate, GCorr 2022 Corporate, GCorr 2021 Corporate, GCorr 2020 Corporate, GCorr 2019 Corporate, and GCorr 2018 Corporate. As the table shows, the R-squared values in the overall sample based on GCorr 2023 Corporate are similar to the values for GCorr 2022 Corporate with slightly lower mean and median. In addition to this, dispersion, as described by standard deviation, is slightly lower in GCorr 2023 Corporate compared to GCorr 2022 Corporate. When comparing GCorr 2023 with the pre-pandemic GCorr 2019 Corporate model, we also observed the mean R-squared values to be slightly lower in GCorr 2023. This decrease is driven by the fact that relatively smaller firms were added to the database in GCorr 2021.

Table 5 R-squared Summary Statistics

MODEL	Number of Firms	MEAN	STD.DEV.	MINIMUM	25TH PCT	MEDIAN	75TH PCT	MAXIMUM
GCorr 2018 Corporate	46,783	21.6%	14.0%	10.0%	10.0%	15.5%	29.5%	65.0%
GCorr 2019 Corporate	46,475	20.8%	13.3%	10.0%	10.0%	15.0%	28.4%	65.0%
GCorr 2020 Corporate	48,218	22.4%	13.1%	10.0%	10.4%	18.9%	30.7%	65.0%
GCorr 2021 Corporate	51,569	20.0%	12.3%	10.0%	10.0%	14.7%	27.1%	65.0%
GCorr 2022 Corporate	52,461	19.5%	11.4%	10.0%	10.0%	15.0%	25.2%	65.0%
GCorr 2023 Corporate	53,436	19.2%	11.4%	10.0%	10.0%	14.7%	25.0%	65.0%

Figure 5 through Figure 8 show the distribution of R-squared values in GCorr 2023 Corporate and GCorr 2022 Corporate for the sample of all public firms in our database, financial firms, industrial firms, and utility firms.

Figure 5 R-squared distribution, GCorr 2023 Corporate and GCorr 2022 Corporate: All firms.



The classification of firms into sectors is carried out as follows:

- A firm belongs to the financial sector if its primary industry is N06, N23, N24, N29, N30, N31, N32, N46, N47, or N48.
- A firm is a utility firm if its primary industry is N58, N59, or N60.
- If a firm does not belong to either of these sectors, it is classified as an industrial firm.

[3]The sample for each of the models is determined by selecting all publicly-traded firms, for which we can calculate an asset return during the last six years. *For example*, there are 53,436 firms with asset returns April 2017 — March 2023. These firms constitute the sample for GCorr 2023 Corporate.

Figure 6 R-squared distribution, GCorr 2023 Corporate and GCorr 2022 Corporate: Financial firms.

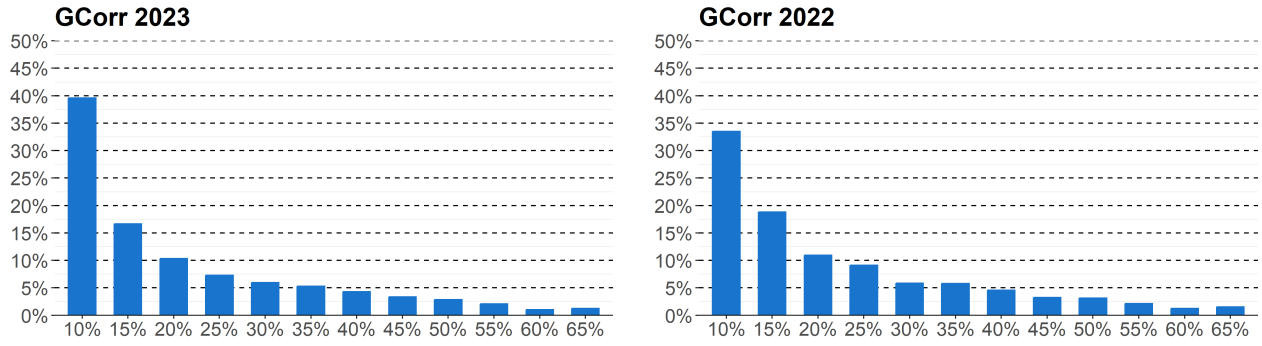


Figure 7 R-squared distribution, GCorr 2023 Corporate and GCorr 2022 Corporate: Industrial firms.

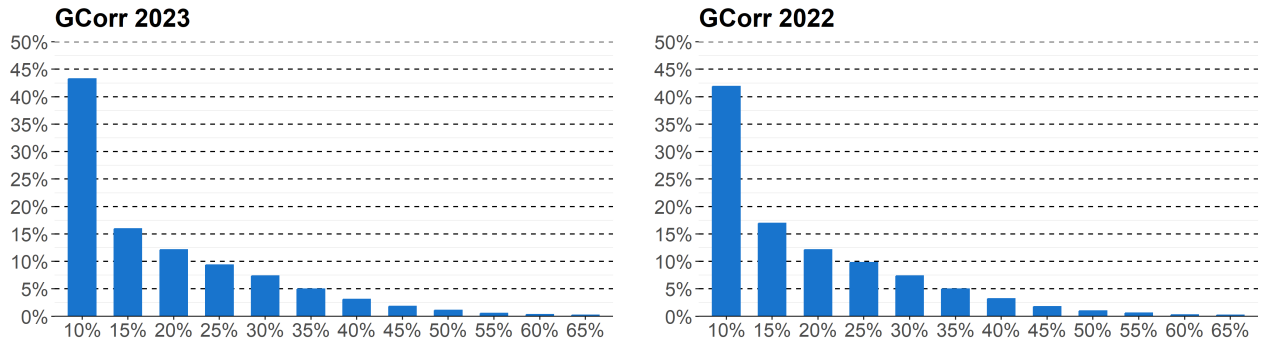
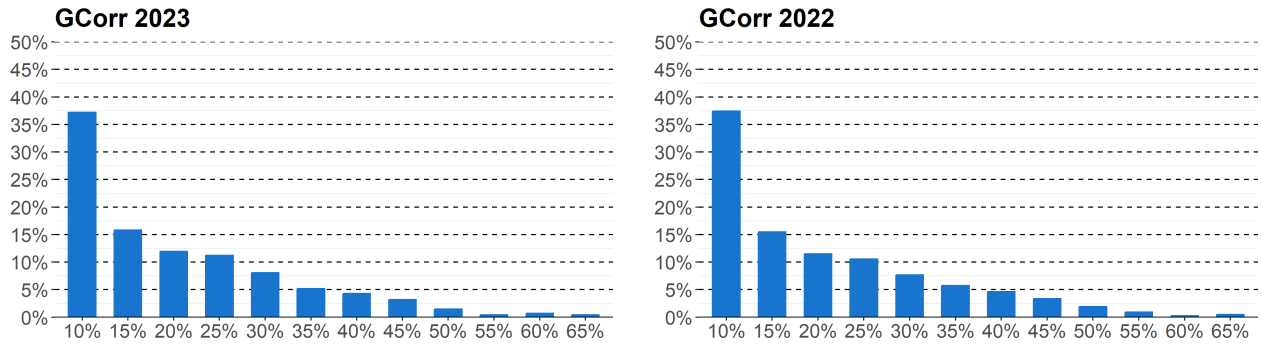


Figure 8 R-squared distribution, GCorr 2023 Corporate and GCorr 2022 Corporate: Utility firms.



R-SQUARED VALUES BY COUNTRY

In this section, we determine the median R-squared values by country. In Figure 9, we present the top-ten countries with the highest median R-squared values in GCorr 2023 Corporate, GCorr 2022 Corporate, GCorr 2021 Corporate, and GCorr 2020 Corporate respectively.

Figure 9 Median R-squared values based on GCorr 2023 Corporate, GCorr 2022 Corporate, GCorr 2021 Corporate, and GCorr 2020 Corporate: Top 10 countries.

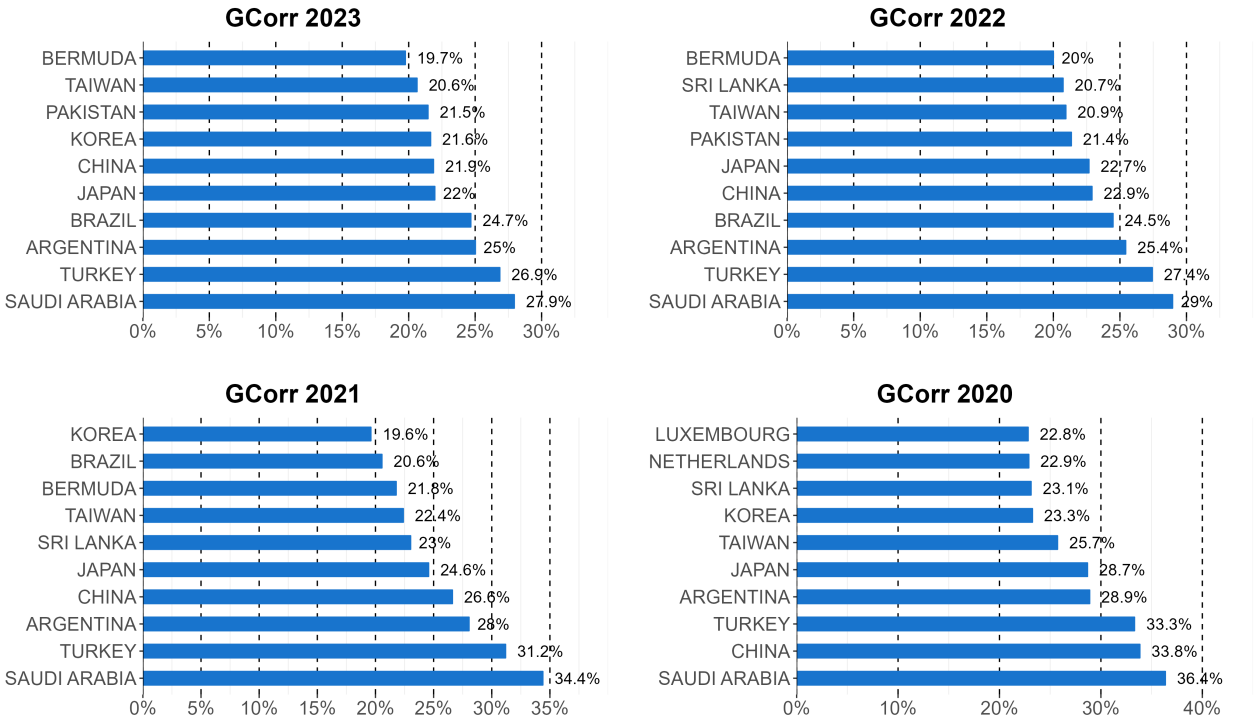
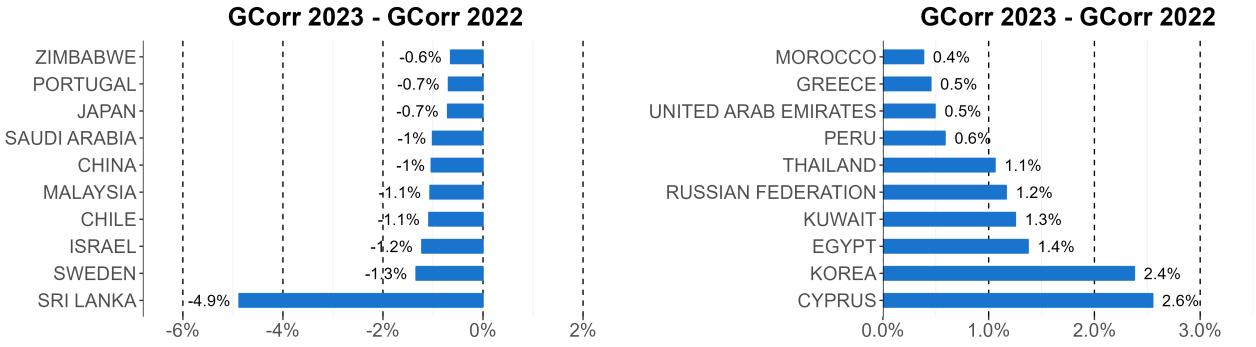


Figure 10 shows the top 10 countries in which we observe the largest increase or decrease in the median level of R-squared from GCorr 2022 Corporate to GCorr 2023 Corporate.

Figure 10 Negative and positive differences in median R-squared values (GCorr 2023 Corporate — GCorr 2022 Corporate): Top 10 countries.



R-SQUARED VALUES BY INDUSTRY

This section presents the median R-squared values by industry. In Figure 11, we list the top 10 industries with the highest median R-squared values in GCorr 2023 Corporate, GCorr 2022 Corporate, GCorr 2021 Corporate, and GCorr 2020 Corporate respectively.

Figure 11 Median R-squared values based on GCorr 2023 Corporate, GCorr 2022 Corporate, GCorr 2021 Corporate, and GCorr 2020 Corporate: Top 10 industries.

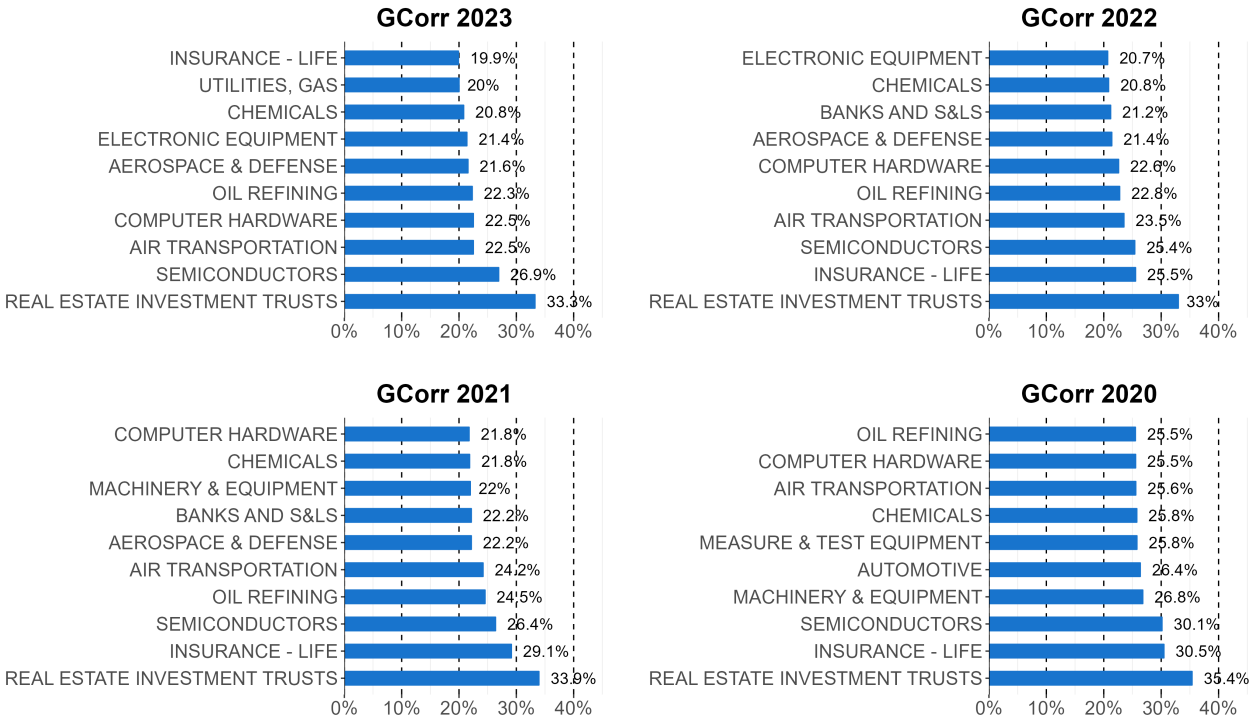
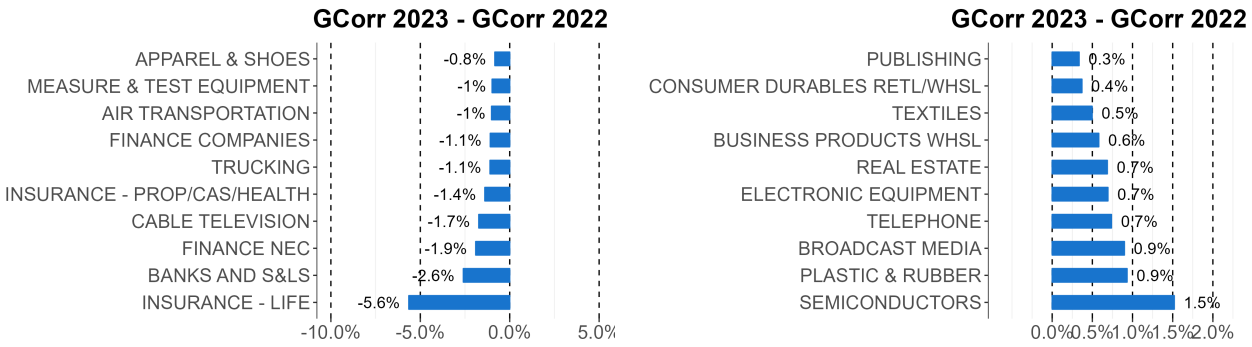


Figure 12 shows the top 10 industries in which we observe the largest increase or decrease in the median level of R-squared from GCorr 2023 Corporate to GCorr 2022 Corporate.

Figure 12 Negative and positive differences in median R-squared values (GCorr 2023 Corporate — GCorr 2022 Corporate): Top 10 industries.

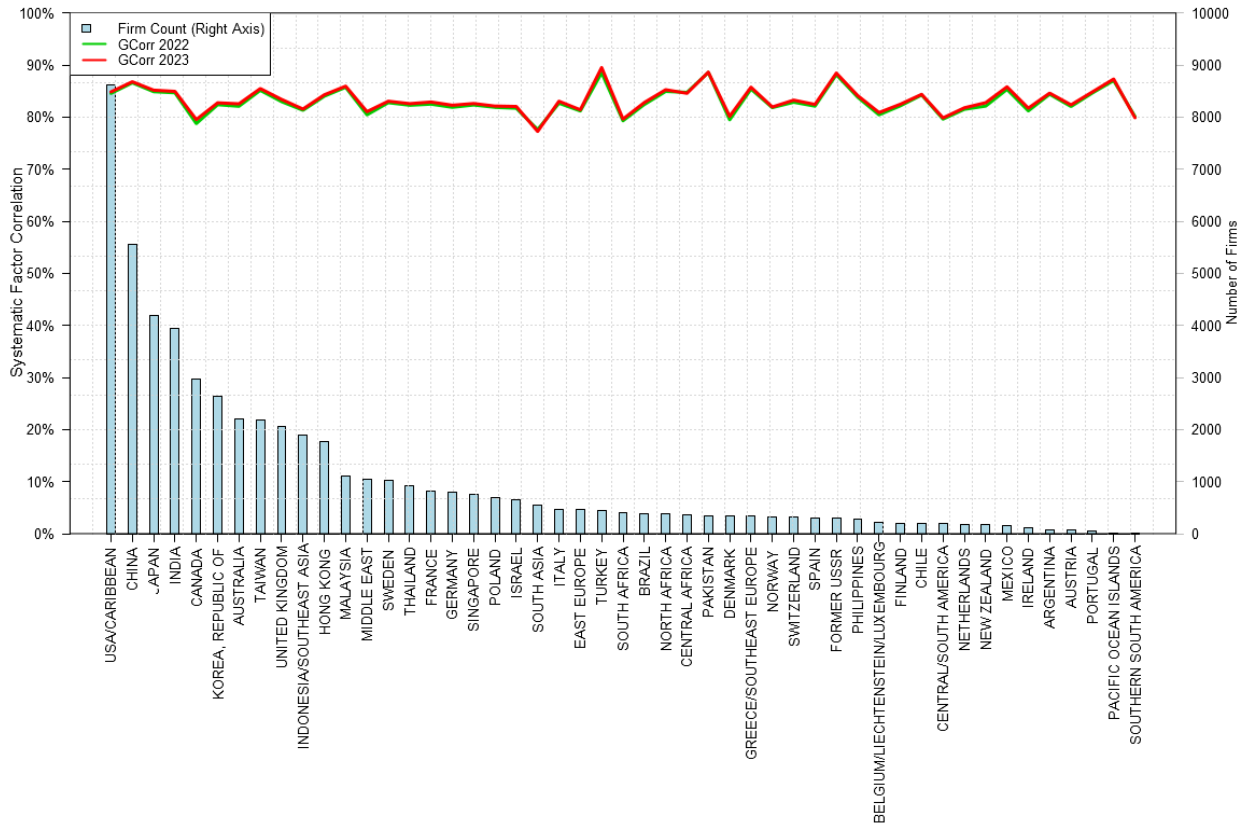


5. Systematic Factor Correlation Analysis

Asset correlations are driven by both R-squared values and the systematic factor correlations. To understand the changes in asset correlations, we must also look at the change in the systematic factor correlations.

To measure the change between GCorr® 2022 and GCorr 2023, we calculate the 61x61 systematic factor correlations within each country and look at the average. Figure 13 shows the average systematic factor correlations within each country under GCorr 2022 and GCorr 2023 for countries in which we have firms.

Figure 13 Average systematic factor correlation under GCorr 2023 Corporate and GCorr 2022 Corporate.



Systematic factor correlations tend to be very stable over time. The country average systematic factor correlations calculated using GCorr 2023 are at the same level or slightly higher than those calculated using GCorr 2022.

6. Portfolio Impact Analysis

In this section, we compare risk measures of 16 portfolios using RiskFrontier under GCorr® 2023 Corporate and GCorr 2022 Corporate correlation models, while keeping other parameters, including default probabilities, constant. The portfolios are designed to represent realistic investments, so we conduct analyses using firm-specific Expected Default Frequency (EDF) values.

We create portfolios covering four regions: Global, United States, Europe, and Asia/Australia. Within each region, we create four portfolios representing different compositions: non-financials, low risk non-financials, financials, and large financials. For each portfolio, we compute the average correlation statistics under GCorr 2022 Corporate and GCorr 2023 Corporate. We then run each portfolio under the two correlation models and compare the unexpected loss and economic capital to show the impact of the updated correlation measures.

In general, the directional change in asset correlations matches the directional change in the unexpected loss and economic capital. Lower asset correlations result in a lower likelihood of realizing extreme losses, which generally decreases the unexpected loss and economic capital. This finding is what we observe in most portfolios. The magnitude of the impact depends on many other factors, such as the EDF level and the level of asset correlations.

Table 6 Portfolio Characteristics

PORTFOLIO	NUMBER OF FIRMS	AVERAGE GCORR 2023 RSQ (DIFFERENCE FROM 2022)	AVERAGE GCORR 2023 ASSET CORRELATION (DIFFERENCE FROM 2022)	GCORR 2023 UNEXPECTED LOSS IN BPS (% CHANGE FROM 2022)	GCORR 2023 ECONOMIC CAPITAL IN BPS (% CHANGE FROM 2022)
Global					
Non-Financials	33,997	19.62% (-0.03%)	7.86% (0.03%)	143.0 (0.28%)	765.0 (-0.20%)
Non-Financials, EDF < 100 bps	16,667	23.02% (0.01%)	9.26% (0.04%)	19.9 (0.51%)	165.0 (0.00%)
Financials	7,413	22.83% (-0.64%)	10.88% (-0.38%)	94.8 (-2.57%)	667.2 (-3.15%)
Financials, Assets > $25 billion	737	38.18% (-0.83%)	19.74% (-0.49%)	120.4 (-2.11%)	992.2 (-3.33%)
U.S.					
Non-Financials	4,187	21.02% (0.12%)	16.16% (0.10%)	421.9 (0.60%)	2062.5 (0.43%)
Non-Financials, EDF < 100 bps	1,440	29.97% (0.38%)	23.79% (0.37%)	41.5 (1.72%)	467.5 (2.25%)
Financials	1,693	25.55% (-1.73%)	20.66% (-1.75%)	165.9 (-4.60%)	1318.5 (-6.00%)
Financials, Assets > $25 billion	142	49.45% (-1.39%)	44.29% (-1.19%)	252.9 (-2.28%)	2430.7 (-4.24%)
Europe					
Non-Financials	5,293	17.60% (-0.16%)	10.06% (-0.10%)	135.4 (-1.31%)	820.3 (-0.12%)
Non-Financials, EDF < 100 bps	2,870	21.06% (-0.03%)	12.16% (-0.03%)	23.2 (-0.43%)	215.2 (0.00%)
Financials	1,558	24.34% (-0.53%)	16.18% (-0.38%)	79.9 (-2.56%)	706.0 (-2.22%)
Financials, Assets > $25 billion	164	38.57% (-0.15%)	26.41% (-0.19%)	100.4 (-1.76%)	988.6 (-1.55%)
Asia/Australia					
Non-Financials	21,236	20.26% (-0.05%)	8.59% (0.02%)	111.5 (0.36%)	651.2 (0.28%)
Non-Financials, EDF < 100 bps	11,073	22.83% (-0.07%)	9.69% (0.01%)	20.9 (0.00%)	173.3 (-1.25%)
Financials	3,117	21.55% (-0.32%)	9.50% (-0.09%)	89.6 (-0.33%)	571.6 (0.09%)
Financials, Assets > $25 billion	345	34.69% (-1.29%)	18.20% (-0.67%)	136.7 (-1.87%)	1013.7 (-2.83%)

7. Summary

Asset correlations play an important role in determining credit portfolio risk. When we release new and updated correlation estimates each year, we provide a summary of changes from the previous release and their potential impact on portfolio risk measures. In February 2024, we released the most recent correlation estimates, which we refer to as GCorr® 2023 Corporate. To estimate these correlations, we used data covering July 1999—March 2023. Compared to GCorr 2022 Corporate, the new estimates use additional asset return data generated from June 2022—March 2023.

This document summarizes the correlation estimates from the Moody's GCorr 2023 Corporate model. We present statistics of the asset correlations, R-squared values, and factor correlations by country, industry, and sample based on firm size. We also compare the estimates from GCorr 2023 Corporate to the previous three models: GCorr 2022 Corporate, GCorr 2021 Corporate, and GCorr 2020 Corporate. In addition, this document discusses the impact of the new estimates on risk measures of 16 representative loan portfolios.

Overall, the average levels of estimated correlations in GCorr 2023 Corporate are very close to but slightly higher than those in GCorr 2022 Corporate. Compared to GCorr 2019, the correlations are similar in GCorr 2023. We expect that using GCorr 2023 Corporate will lead to varying risk measures compared to GCorr 2022 Corporate based on portfolio composition.

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